

MD KHALEQUZZAMAN CHOWDHURY SAYEM

3D AI Engineer, LiOPS, Seoul, South Korea

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Research Summary

3D AI Engineer at **LiOPS**, working on stereo depth estimation and 3D perception for industrial systems, with a research background in **3D vision and vision-language models**. At UNIST, I worked on real-time two-hand manipulation modeling (**AAAI 2025**), **HandVQA (CVPR 2026)**, a **1.6M-scale** benchmark for spatial reasoning about hands in VLMs, and 3D Gaussian human avatars (**ECCV 2026**). I am interested in models that understand people: how they move, interact with objects, and behave over time.

Research Interests

3D Computer Vision; Human Behavior Understanding; Human-Object Interaction; Vision-Language Models; World Models; Embodied AI.

Education

M.S. in Computer Science and Engineering

Sep 2023 – Aug 2025

Ulsan National Institute of Science & Technology (UNIST)

Ulsan, Korea

– GPA: 3.93/4.3 (96.30/100)

– **Master's Thesis (Accepted at CVPR 2026):** *HandVQA: Diagnosing and Improving Fine-Grained Spatial Reasoning about Hands in Vision-Language Models.*

– Advisors: Prof. Seungryul Baek (Supervisor), Prof. Binod Bhattarai (Co-supervisor)

B.S. in Industrial Engineering

Sep 2019 – Aug 2023

Ulsan National Institute of Science & Technology (UNIST)

Ulsan, Korea

Conference Publications (* equal contribution)

- **[CVPR 2026]** MD K. C. Sayem*, M. T. Chowdhury*, Y. Y. Tiruneh, M. A. Khan, M. S. Ali, B. Bhattarai, S. Baek. *HandVQA: Diagnosing and Improving Fine-Grained Spatial Reasoning about Hands in Vision-Language Models.*
Project Page: kcsayem.github.io/handvqa/
 - Proposed **HandVQA**, a **1.6M-scale** benchmark derived deterministically from **3D hand pose annotations** across FreiHAND, InterHand2.6M, and FPHA.
 - Diagnosed systematic failures of large VLMs (e.g., LLaVA, Qwen2.5VL) in fine-grained spatial reasoning, revealing depth ambiguity, left/right confusion, and relational bias.
 - Demonstrated that injecting **structured 3D supervision** improves zero-shot transfer across tasks, boosting gesture recognition by **+10.33%** and hand-object interaction recognition by **+2.63%**.
 - Provides a scalable pathway for injecting explicit **3D geometric priors** into text-first foundation models.
- **[ECCV 2026]** Y. Y. Tiruneh, M. S. Ali, U. Jeong, M. A. Khan, MD K. C. Sayem, A. Bayramgeldiyev, B. Bhattarai, S. Baek.
FlexiAvatar: Unified 3D Gaussian Human Avatars Under Arbitrary Body Visibility.
Project Page: yihalem1.github.io/FlexiAvatar/
 - Visibility-aware framework for animatable **3D Gaussian human avatars** from full-body, upper-body, or head-only monocular video.
- **[AAAI 2025]** E. Ismayilzada*, MD K. C. Sayem*, Y. Y. Tiruneh, M. T. Chowdhury, M. Boboev, S. Baek.
QORT-Former: Query-optimized Real-time Transformer for Understanding Two Hands Manipulating Objects.
Project Page: kcsayem.github.io/QORT-Former/
 - Designed the core query optimization method and integrated structured contact-map features into the Transformer architecture.
 - Proposed a **query division strategy** that separates queries into left-hand, right-hand, and object groups via a dedicated query proposal module, reducing ambiguity between symmetric hand features.
 - Reduced Transformer complexity from **300 queries / 6 decoders** to **108 queries / 1 decoder**, achieving **53.5 FPS** real-time inference while improving state-of-the-art accuracy by **+27.2%** (H2O) and **+10.4%** (FPHA).

Industry Experience

LiOPS

3D AI Engineer

May 2026 – Present

Seoul, Korea

- Develop **stereo depth estimation** models for industrial 3D vision systems that require metric accuracy.
- Work on multi-camera calibration, LiDAR and **point-cloud** processing, and synthetic data generation.
- Optimize models for real-time deployment on edge hardware (ONNX / TensorRT, Jetson).

Research Experience

Vision & Learning Lab, UNIST

Researcher

Jan 2023 – Apr 2026

Ulsan, Korea

Researcher (Sep 2025 – Apr 2026); Graduate Research Assistant (Sep 2023 – Aug 2025); Undergraduate Research Assistant (Jan 2023 – Aug 2023)

- Designed geometry-aware Transformer architectures for real-time 3D hand-object interaction modeling, integrating structured contact and pose representations (AAAI 2025).
- Led large-scale experimentation across FreiHAND, InterHand2.6M, and FPHA to analyze cross-view generalization, occlusion robustness, and representation bias in multimodal systems.
- Developed deterministic data generation and diagnostic evaluation pipelines for fine-grained spatial reasoning in foundation VLMs (CVPR 2026).
- Contributed to 3D Gaussian human avatar reconstruction under partial body visibility (ECCV 2026).

Machine Learning & Vision Lab, UNIST

Undergraduate Research Assistant

Sep 2021 – Feb 2022

Ulsan, Korea

- Implemented data augmentation and training improvements for 2D human pose estimation, improving robustness on COCO-style benchmarks.

Selected Industry Research Project

Worker Safety System for Industrial Environments Using Multi-Task YOLOv11

Feb 2025 - Apr 2025

Hyundai Motors & Vision and Learning Lab (UNIST) | [GitHub](#)

- Designed and implemented a unified multi-task perception system performing detection, instance segmentation, and human pose estimation.
- Integrated YOLOv11 detection, segmentation, and pose heads into a single pipeline, reducing inference latency by **35%**.

Technical Skills

- **Core:** PyTorch, CUDA, Distributed/Accelerated Training, Linux, Git, LaTeX
- **Multimodal / Foundation Models:** HuggingFace Transformers, LoRA fine-tuning, VLM benchmarking (e.g., LLaVA, Qwen-VL)
- **3D Vision:** Pose estimation, multi-view geometry, human/hand modeling, stereo depth, point clouds and LiDAR, multi-camera calibration, NeRF / Gaussian Splatting, ONNX / TensorRT
- **Data & Tooling:** NumPy, Pandas, OpenCV, experiment scripting and evaluation pipelines

Teaching Experience

- **Teaching Assistant:** Introduction to AI Programming in Python (Fall 23, Spring 24, Fall 24) and Discrete Mathematics (Fall 24) with Prof. Michael Jon Burrell; Advanced Computer Vision (Spring 24) with Prof. Seungryul Baek.

Leadership

PAIS: Partnership for AI and Society

President

Jan 2021 – Aug 2025

UNIST

- Founded and led an AI-safety discussion group with faculty mentorship, organizing talks and forums on data privacy and responsible AI at UNIST.

Honors & Awards

- **Graduate School Scholarship (Korean Government):** full tuition + stipend for M.S.
- **UNIST Dream Scholarship (UNIST):** full tuition + stipend for B.S.
- **AI Challengers Program (UNIST Innovative Education Center):** \$5K funding for parallelized bandit algorithms.